

Romit Sen

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WORK EXPERIENCE

Applied AI Engineer

July 2026 – Present

MindSpec, Inc.

McLean, VA

- Rebuilt a scientific extraction pipeline around Claude structured outputs and Pydantic validation, populating a 167-field, 9-table evidence schema across 10+ gold-standard publications and reducing per-paper curation from ~3 hours to ~45 minutes plus reviewer sign-off.
- Designed hybrid PostgreSQL/pgvector retrieval using BGE-M3, HNSW, full-text search, and Reciprocal Rank Fusion (RRF), with automated vocabulary checks, provenance-aware deduplication, LLM-as-judge auditing, regression fixtures, and LangSmith observability.

AI Research Intern, MindSpec, Inc.

September 2025 – July 2026

- Developed the v1 RAG pipeline using Docling, PubMedBERT, pgvector, and Claude to convert 50+ biomedical PDFs into 4,400 structured records, including confidence intervals and adjustment metadata, with a 2-pass fallback for context limits.
- Built ontology normalization for exposure terms using exact, token, and RapidFuzz matching, achieving ~91% automatic canonicalization while routing unresolved terms to expert review and linking factors to ChEBI, ECTO, and ICD-10.

Data Analyst Intern

May 2024 – August 2024

Bear Cognition, Inc.

North Charleston, SC

- Replaced manual JotForm workflows with 9 Plotly Dash applications writing directly to Snowflake for a training facility serving Olympic and professional athletes; applications remained in production 12+ months after deployment.
- Prototyped a reusable OCR-based ingestion and analytics tool for LTL shipping companies, parsing receipts and orders into a Dash AG Grid dashboard backed by MongoDB.

Quality Assurance Analyst

July 2021 – September 2022

Tata Consultancy Services

Kolkata, India

- Tested SAP CRM/FI workflows for a Big Four professional services client, logging and triaging 400+ JIRA defects and building automated regression suites with Selenium and Robot Framework.

EDUCATION

Oklahoma State University

Stillwater, OK

M.S. in Business Analytics & Data Science, GPA 4.0/4.0 | CIE Scholar (NSF I-Corps)

August 2023 – May 2025

- Graduate Teaching Assistant, Venture Capital
- Relevant Coursework: Statistics, Descriptive Analytics, Predictive Analytics, Advanced Data Wrangling

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Electronics and Communication Engineering, GPA: 3.6/4.0

July 2017 – July 2021

PROJECTS

Agentic AI HR Automation System | Python, Claude, FastMCP

July 2025

- Built an MCP server (FastMCP) exposing 13 HR tools across onboarding, leave, ticketing, meetings, and email; a single natural-language prompt in Claude Desktop creates the employee record, opens an IT ticket, sends the welcome email, and schedules the kickoff meeting.

E-commerce QA Chatbot with Intent Routing | Python, SQL, RAG

July 2025

- Developed a chatbot that routes queries by intent: text-to-SQL over a live product database for inventory questions and RAG over FAQ documents for policy questions, both on Llama 3.3 via Groq, with a web-scraping pipeline that builds the product catalog.

TECHNICAL SKILLS

Languages: Python (scikit-learn, pandas, NumPy), SQL, R

AI/ML: RAG, LangChain, LangSmith, Claude API, FastMCP, Pydantic, embeddings, pgvector, ChromaDB, Hugging Face

Experimentation & Statistics: Hypothesis Testing, A/B Testing, Multivariate Analysis

Visualization: Power BI, Tableau, Plotly Dash, Looker, Streamlit